

Logic Gates · AS · Answers

24 written + 6 pen-first = 30 · Student copy

One mark per point unless grouped. Accept equivalent clear wording. Do not credit contradictory answers. Authored mark groupings are specified below; all rehearsal specimens are freshly authored.

1. Expression table [4]

NOT A: **11110000**; B OR C: **01110111**; X: **01110000**. Award one mark for each consecutive correct pair in X: 01 / 11 / 00 / 00. Working columns are optional and carry no separate marks. This grouping is our authored scheme.

2. Name the behaviour [2]

(a) **NAND** [1]. (b) **NOR** [1].

3. OR and XOR [2]

A=1, B=1 [1]; OR=1 and XOR=0, both outputs [1].

4. Read the circuit [4]

(a) **X = NOT(A AND B) AND (B OR C)**. Accept (A NAND B) AND (B OR C). NOT(A AND B) / A NAND B [1]; B OR C [1]; both combined by AND [1]. Preserve the drawing's structure; NOT(A AND B) is not (NOT A) AND B. (b) The bubble **inverts/negates** the gate's output [1].

5. Equivalent functions? [4]

(a) At 000, X = 1 AND 0 = **0** [1]; Y = NOT 0 = **1** [1]. (b) **Not equivalent** [1]; equivalent functions must agree on every input combination, and this is a counterexample [1]. "They look different" is NE.

6. Authored expression rehearsal [2]

A NOR B = 11000000; XOR with C = 01010101 gives X = **10010101**. Our complete-half scheme: first four rows **1001** [1]; last four **0101** [1]. No separate intermediate-column points.

7. Status light [6]

(a) **L = A XOR B**: XOR [1], correctly applied to A and B [1]. Accept $A \oplus B$. (b) Output **0110**; one mark for each consecutive correct pair: 01 / 10 [2]. (c) **11** [1]; OR incorrectly gives 1, required output 0, both [1].

8. Mystery box - pen first [6]

(a) NAND [1]; A=1 and B=1 / 11 [1]. (b) A=0 gives 1 [1]; A=1 gives 0 [1]. (c) NOT / inversion [1]; these are all possible values of a single binary input and the output is its opposite in both cases [1]. Merely naming NOT twice earns only the name mark. Human marked; running the model is a check, not another mark.

Extension, unmarked: let $T = \text{NAND}(A,B)$, then $\text{NAND}(T,T) = A \text{ AND } B$. The second gate inverts the first result. This proves that NAND is sufficient to build AND, not that manufacturers use only NAND. Real cell libraries contain multiple functions. Karel 09/10 is outside the total.